Do Latent Vector Fields Align?

How angle-preserving alignment breaks down under autoencoder dynamics

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Abstract

The Platonic Representation Hypothesis suggests that independently trained models converge toward a shared representation. Separately, an autoencoder defines a latent vector field through iterated encoding–decoding. We ask whether these latent vector fields align across models differing only in random seed. They do not: the alignment that holds for the base embeddings erodes as the field is iterated. Two models can look nearly aligned by cosine, yet their dynamics drive points to model-specific destinations. Latent vector fields are therefore a weaker basis for comparing models than the static representations.

https://github.com/LeonardoSani/LAAB

1. Introduction

Relative representations (Moschella et al., 2023) encode a point by its cosine similarities to shared reference *anchors*, giving a code invariant to angle-preserving maps and reproducible across independently trained models, in line with the Platonic Representation Hypothesis (Huh et al., 2024). Separately, an autoencoder's latent self-map $f = E \circ D$ induces a latent vector field (Fumero et al., 2026) whose iterated dynamics drive points to attractors.

(Fumero et al., 2026) leaves open a question: *are the latent vector fields of independently trained networks aligned?* A positive answer would give an alignment tool complementary to the static embeddings; we give a first, negative one for convolutional autoencoders. **Our contributions:**

- **Dynamic relative representations.** We extend relative representations (Moschella et al., 2023) from static embeddings to latent dynamics.
- **Alignment decays with the dynamics.** Aligned at the base, then decaying as the dynamics evolve.
- **The decay is structured.** Some points' orbits diverge across seeds more than others within the uniform collapse.

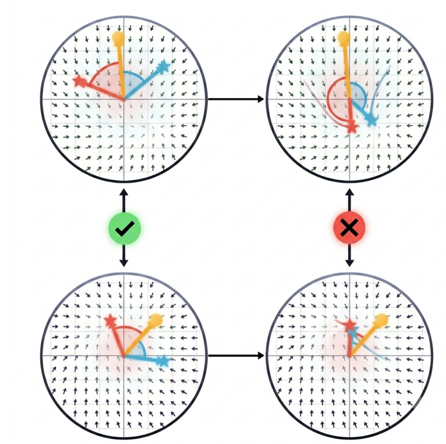


Figure 1. **Qualitative sketch.** Two latent vector fields, aligned at the base, drift apart as the field is iterated toward its attractors.

- **The decay has downstream cost.** It corrupts cross-model stitching.
- **The cause is provable.** Attractor alignment requires shared basins, which independent training does not produce.

2. Related work

Comparing representations. Much work asks whether independently trained models learn the same representation, via similarity indices like CKA (Kornblith et al., 2019) or, more strongly, the Platonic Representation Hypothesis (Huh et al., 2024). **Relative representations** (Moschella et al., 2023) make this operational with a cross-encoder-invariant code that enables zero-shot latent communication; **model stitching** tests the same equivalence functionally, splicing one model's features into another (Lenc and Vedaldi, 2015; Bansal et al., 2021). All of these compare *static* features. **Latent dynamics.** (Fumero et al., 2026) instead reads the autoencoder as a latent vector field with attractors, and asks whether these fields align across models. We extend the static relative representation into two *dynamic* ones, using zero-shot stitching (Moschella et al., 2023) as functional evidence of the breakdown.

3. Method

Notation and background. We take a family $\{F_s = D_s \circ E_s\}_{s \in \mathcal{S}}$, $E : \mathcal{X} \to \mathcal{Z}$, $D : \mathcal{Z} \to \mathcal{X}$, identical up to the seed s, one as reference (Appendix C). Encoder and decoder compose into the **latent self-map** $f_s = E_s \circ D_s$. Following (Fumero et al., 2026), f_s defines a **latent vector field** $V_s(z) = f_s(z) - z$; we study its dynamics through the depth-t trajectory

$$\phi_s^0(x) = E_s(x), \qquad \phi_s^{t+1}(x) = f_s\big(\phi_s^t(x)\big),$$

with ϕ_s^∞ the attractor at convergence; we sweep depths $\mathcal{T} = \{0, 1, \ldots, \infty\}$.

Dynamic and static relative representations. Fix an **anchor set** $\mathcal{A} = \{x_i\}_{i=1}^N$, shared across seeds. The static **Base** representation $\mathcal{B}_s(\cdot)$ is

$$\mathcal{B}_s(x)_i = \cos\big(E_s(x),\, E_s(x_i)\big), \qquad i = 1, \ldots, N,$$

reproducible across seeds up to an angle-preserving T_0, $\mathcal{B}_{s'} \approx \mathcal{B}_s$ (Moschella et al., 2023). Iterating the anchors, and optionally x, gives its dynamic counterparts **Mixed** $\mathcal{M}_s^t(\cdot)$ and **Full** $\mathcal{F}_s^t(\cdot)$,

$$\mathcal{M}_s^t(x)_i = \cos\big(E_s(x),\, \phi_s^t(x_i)\big), \qquad i = 1, \ldots, N,$$

$$\mathcal{F}_s^t(x)_i = \cos\big(\phi_s^t(x),\, \phi_s^t(x_i)\big), \qquad i = 1, \ldots, N,$$

both equal to $\mathcal{B}_s$ at $t = 0$: Mixed asks whether the fixed T_0 still relates cross-seed dynamics, Full whether any T_t (in general different from T_0) does (Appendix A).

Metrics. For a pair $\{s, s'\}$ and input x, write $r_s, r_{s'} \in \mathbb{R}^N$ for the two relative representations. We use:

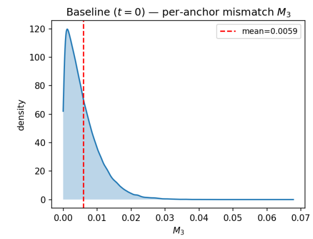
$$M_1(x; s, s') = \|r_s - r_{s'}\|_2, \quad M_2(x; s, s') = 1 - \cos\big(r_s, r_{s'}\big),$$

$$M_3(x; s, s')_i = \big|r_{s,i} - r_{s',i}\big|, \quad M_4(x; s) = \big\|x - D_r(r_s)\big\|_2^2,$$

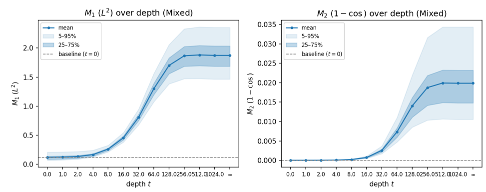
with M_4 the stitching error of decoding s through a relative decoder D_r trained on the reference, averaged over pixels, yielding the MSE, and per-anchor badness $\mu_i = \mathbb{E}_{\{s,s'\}} \mathbb{E}_x\, M_3(x; s, s')_i$. We track these across $\mathcal{T}$, pooling M_1, M_2, M_3 over points and all $\binom{|\mathcal{S}|}{2}$ pairs and averaging M_4 over points and non-reference seeds. At $t = \infty$ we add **basin consistency** J_i, the cross-seed Jaccard of each anchor's attractor basin (Appendices A and C), which we use to explain the alignment decay.

4. Results

A high-cosine breakdown. The codes should reproduce across seeds, and at $t = 0$ they do, all metrics at a common floor (Fig. 2a; the regime of (Moschella et al., 2023)); we measure how far they drift under iteration. The scale-invariant gauge is the cross-seed angle between codes: negligible at the base ($0.46°$, $\cos \approx 1$), it widens under Mixed iteration ($\mathcal{M}^t$) to $11.5°$ at $t = \infty$ ($\cos \approx 0.98$, $M_2 \approx 0.020$; Fig. 2b), a $\sim 25\times$ drift. The absolute 0.98 looks aligned, but the reference is the base's ≈ 1, against which agreement has collapsed. The Euclidean distance $M_1 \approx 1.87$ agrees but is set by the uninformative code norm (Appendix B), so we read the breakdown from the angle M_2 and the per-anchor structure.



(a) At $t = 0$, M_3 near zero.



(b) Mixed M_1, M_2 over depth.

Figure 2. Aligned at the base, drifting under iteration. At $t = 0$ the relative code reproduces across seeds ($M_3 \approx 0$, a); once the anchors are iterated (Mixed), the cross-seed gap M_1, M_2 grows with depth and saturates by $t = \infty$ (b).

Per-anchor structure. M_3 decomposes the L^2 gap across anchors, broadening into a heavy right tail as the field iterates (Figs. 3, 4): each anchor becomes a less reliable model-independent ("Platonic") reference. The collapse is uniform—even the best attractor anchor is far worse than any baseline anchor. Yet it is ordered: across independent seed pairs some anchors stay consistently worse and others less so (Fig. 5), and the same structure appears across digit classes (Fig. 6).

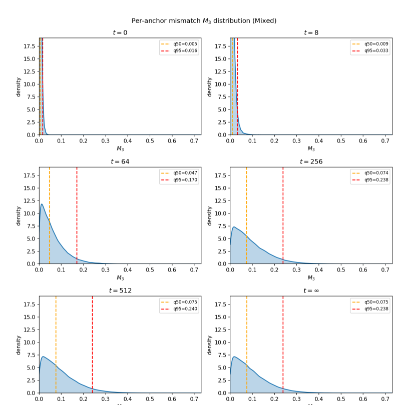


Figure 3. M_3 broadens into a heavy right tail under iteration: every anchor becomes less reproducible across seeds, but some fail much more strongly.

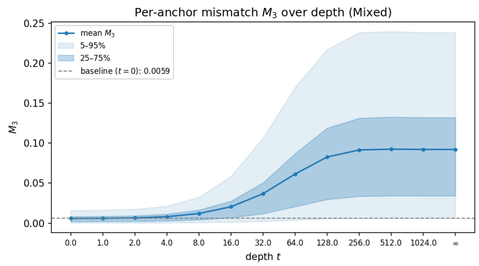


Figure 4. The per-anchor mismatch M_3 grows with depth (mean $0.006 \to 0.092$), with high-percentile anchors drifting fastest.

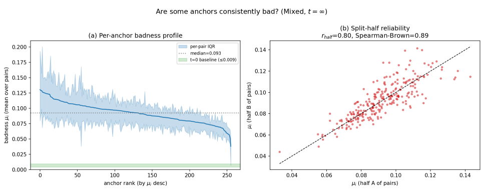


Figure 5. The good/bad ordering is reproducible. Per-anchor badness μ_i with its per-pair IQR (a), and its split-half reliability ($r = 0.80$, Spearman–Brown 0.89, b): the same anchors stay worse across independent seed pairs—a stable ordering within the otherwise uniform collapse.

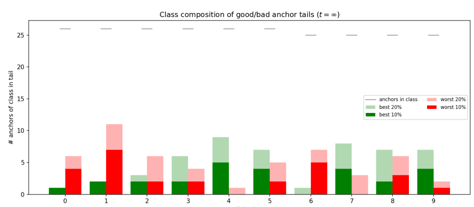


Figure 6. The worst anchors cluster by digit class. Class composition of the worst/best M_3 tails: **1** and **6** dominate the worst tail, while **4** and **7** never enter it.

Stitching. This drift carries an operational cost. When one seed's attractor-anchor code is decoded zero-shot through the reference's relative decoder, reconstruction error rises $\sim 12\times$ from $t = 0$ to $t = \infty$ (Fig. 7; qualitative reconstructions in Appendix 5). The codes drift enough to corrupt reconstruction though cosine still reads them aligned.

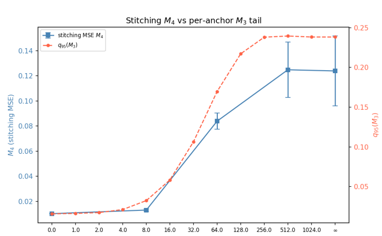


Figure 7. The drift has an operational cost. Zero-shot stitching error M_4 rises $\sim 12\times$ from $t = 0$ to $t = \infty$, tracking the per-anchor tail M_3^{q95}.

Full representation and its cause. Iterating the test points too (Full $\mathcal{F}^t$) widens the gap further ($\cos \approx 0.96$, $M_1 = 2.81$ at $t = \infty$, worse than Mixed; Fig. 8); as the most permissive probe—it admits *any* angle-preserving T—none realigns the iterated fields. The cause is the attractor basins: they are not shared across seeds (mean $J_i = 0.025$ matches the null 0.019, i.e. at chance; Fig. 9), and shared basins are necessary for Full alignment (Appendix A).

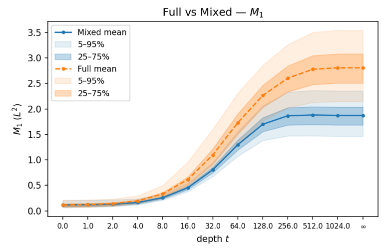


Figure 8. Even the most permissive probe fails. Iterating the test points too (Full $\mathcal{F}^t$) widens the cross-seed gap beyond Mixed ($M_1 = 2.81$, $M_2 = 0.040$ at $t = \infty$): no angle-preserving map realigns the fully iterated fields.

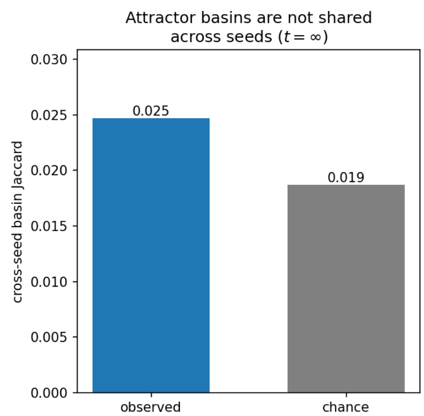


Figure 9. The cause: basins are not shared. At $t = \infty$, the per-anchor cross-seed basin overlap (Jaccard) sits at its size-matched chance level (0.025 vs 0.019): independently trained fields do not share their attractor basins—which Appendix A proves necessary for alignment.

5. Conclusions and discussion

The latent vector fields of independently trained autoencoders do not align under any angle-preserving map. The Platonic convergence holds for base embeddings but not dynamics: each model flows to its own attractors. Alignment decays with depth and, at the attractors, fails: it requires shared basins, which independent training does not produce (Appendix A). Latent dynamics are a weaker basis for comparison than the static representations. Aligning them—if possible at all—requires transformations beyond angle-preserving maps. We show this for MNIST and convolutional autoencoders only; generalization and why anchors fail worst remain open (Appendix D).

References

Yamini Bansal, Preetum Nakkiran, and Boaz Barak. Revisiting model stitching to compare neural representations. In *NeurIPS*, 2021.

Marco Fumero, Luca Moschella, Emanuele Rodolà, and Francesco Locatello. Navigating the latent space dynamics of neural models. In *ICLR*, 2026.

Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. The platonic representation hypothesis. In *ICML*, 2024.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *ICML*, 2019.

Karel Lenc and Andrea Vedaldi. Understanding image representations by measuring their equivariance and equivalence. In *CVPR*, 2015.

Luca Moschella, Valentino Maiorca, Marco Fumero, Antonio Norelli, Francesco Locatello, and Emanuele Rodolà. Relative representations enable zero-shot latent space communication. In *ICLR*, 2023.

Appendix

Appendix A: Mixed, Full, and a necessary condition for alignment

Representations and transformations. Each of the Base, Mixed, and Full codes (§ Method) makes cross-seed agreement equivalent to a transformation relating the iterated maps ϕ_s^t (with $\phi_s^0 = E_s$):

$$\begin{aligned} \mathcal{B}_{s'} \approx \mathcal{B}_s &\iff E_{s'} \approx T_0\, E_s, && \text{(confirmed)} \\ \mathcal{M}_{s'}^t \approx \mathcal{M}_s^t &\iff \phi_{s'}^t \approx T_0\, \phi_s^t, && \text{(if aligned)} \\ \mathcal{F}_{s'}^t \approx \mathcal{F}_s^t &\iff \phi_{s'}^t \approx T_t\, \phi_s^t. && \text{(if aligned)} \end{aligned}$$

The first line holds (Moschella et al., 2023); the other two are conditions, not claims—cross-seed agreement *would* force such a T, which we test. Mixed leaves the input x at the base, so the *same* T_0 must carry over to the iterated anchors; Full iterates the input as well, allowing a possibly different T_t at each depth (in general $\neq T_0$)—the most permissive probe.

Basins of attraction. For one seed, (Fumero et al., 2026) defines an attractor's basin in the *latent* space,

$$B_s(z^*) = \{\, z_0 \in \mathcal{Z}_s : f_s^\infty(z_0) = z^* \,\}.$$

Relative representations see attractors only through cosine, so two with $\cos = 1$ are indistinguishable—equal, and equally informative. We merge their basins, replacing exact with cosine equality:

$$\hat{B}_s(z^*) = \{\, z_0 \in \mathcal{Z}_s : \cos(f_s^\infty(z_0), z^*) = 1 \,\} = \bigcup_{\cos(z', z^*) = 1} B_s(z').$$

The $\hat{B}_s$ still live in the unshared latent space $\mathcal{Z}_s$, so they are not comparable across seeds; we pull them back through the encoder to the shared data. The *directional pre-image* of a point x is

$$P_s(x) = E_s^{-1}\big(\hat{B}_s(\phi_s^\infty(x))\big) \cap \mathcal{D} = \{\, x' \in \mathcal{D} : \cos(\phi_s^\infty(x'), \phi_s^\infty(x)) = 1 \,\}.$$

Applied to anchor x_i, $P_s(x_i) = \{x \in \mathcal{D} : \mathcal{F}_s^\infty(x)_i = 1\}$ is its *attractor basin*—what we mean throughout by an attractor anchor's basin.

Proposition 1. *If $\mathcal{F}_s^\infty \equiv \mathcal{F}_{s'}^\infty$ on $\mathcal{D}$, then $P_s(x_i) = P_{s'}(x_i)$ for every anchor i. Agreement of the directional pre-images is therefore* necessary *for Full alignment.*

Proof. Each pre-image is the level set of a single coordinate, $P_s(x_i) = \{x \in \mathcal{D} : \mathcal{F}_s^\infty(x)_i = 1\}$. Since $\mathcal{F}_s^\infty \equiv \mathcal{F}_{s'}^\infty$, for every $x \in \mathcal{D}$ and every anchor i,

$$x \in P_s(x_i) \iff \mathcal{F}_s^\infty(x)_i = 1 \iff \mathcal{F}_{s'}^\infty(x)_i = 1 \iff x \in P_{s'}(x_i),$$

hence $P_s(x_i) = P_{s'}(x_i)$. □

Necessary, not sufficient. Pre-image agreement, $P_s(x_i) = P_{s'}(x_i)$, is necessary but not sufficient: it matches only how inputs partition into basins, not where each attractor points.

Estimator. We relax exact equality $\cos = 1$ to a tolerance $\tau = 0.99$, $P_s(x_i) = \{x \in \mathcal{D} : \cos(\phi_s^\infty(x), \phi_s^\infty(x_i)) > \tau\}$; the consistency J_i (§ Method) is the mean per-pair Jaccard of these sets, and J its anchor mean,

$$J_i = \frac{1}{\binom{|\mathcal{S}|}{2}} \sum_{s<s'} \frac{\big|P_s(x_i) \cap P_{s'}(x_i)\big|}{\big|P_s(x_i) \cup P_{s'}(x_i)\big|}. \tag{1}$$

Collapsed attractors yield large basins overlapping by construction; we reference a size-matched null with independent membership ($a = |P_s(x_i)|$, $b = |P_{s'}(x_i)|$, $n = |\mathcal{D}|$),

$$J_i^{\text{null}} = \frac{1}{\binom{|\mathcal{S}|}{2}} \sum_{s<s'} \frac{ab/n}{a + b - ab/n}. \tag{2}$$

The necessary condition is not met: $J \approx J^{\text{null}}$, at chance (Fig. 9), so by the Proposition Full alignment is precluded at $t = \infty$—and, Full being the most permissive probe (any T_t versus Mixed's fixed T_0), every angle-preserving map with it, Mixed included. The gap is not even an anchor-selection artifact: extending the reference set from $\{x_i\}$ to all of $\mathcal{D}$ leaves it unchanged.

Appendix B: Cosine and L^2 are the same drift, scaled by the code norm

M_1 and M_2 are the same drift, up to the code norm. Expanding the squared L^2 distance of two codes $r_s, r_{s'} \in \mathbb{R}^N$ with $\cos(r_s, r_{s'}) = 1 - M_2$ gives

$$M_1^2 = \underbrace{\big(\|r_s\| - \|r_{s'}\|\big)^2}_{\text{norm difference}} + \underbrace{2\|r_s\|\|r_{s'}\|\, M_2}_{\text{direction}}. \tag{3}$$

As t grows, the norm difference is negligible beside the direction term (Table 1) so with $\|r_s\| \approx \|r_{s'}\| \equiv R$,

$$M_1^2 \approx 2R^2 M_2. \tag{4}$$

The large $M_1 \approx 1.87$ is thus the small $M_2 \approx 0.020$ amplified by the code norm ($R \approx 9.5$), carrying nothing beyond it. We therefore read the breakdown from the scale-invariant M_2 and its per-anchor split M_3, not from M_1.

Table 1. Decomposition of M_1^2 (Eq. (3)) across depth, pooled over all test points and seed pairs. M_1 is the mean distance and M_1^2 the mean squared distance; $|\Delta\|r\||$ is the mean absolute cross-seed norm gap; the direction term dominates once the codes drift.

t	$\|r\|$	M_1	M_1^2	$\|\Delta\|r\|\|$	M_2	direction
0	11.84	0.12	0.0156	0.065	3.2×10^{-5}	55.1%
8	11.68	0.26	0.0685	0.072	2.2×10^{-4}	87.9%
64	10.77	1.30	1.7082	0.120	7.3×10^{-3}	98.7%
512	9.53	1.88	3.6114	0.231	1.99×10^{-2}	97.7%
∞	9.50	1.87	3.5790	0.237	1.99×10^{-2}	97.5%

Appendix C: Implementation details

Dataset. MNIST. The evaluation set $\mathcal{D}$ is the full test split, $|\mathcal{D}| = 10{,}000$ images, disjoint from training.

Models. $|\mathcal{S}| = 10$ convolutional autoencoders, one per seed $s \in \mathcal{S} = \{1, \dots, 10\}$ ($s = 1$ the reference), with the architecture of (Fumero et al., 2026) (their Table 4): a four-layer Conv2d encoder (channels $d \to 2d \to 4d \to 8d$, 3×3, stride 2, ReLU) with a linear bottleneck, and a mirrored ConvTranspose2d decoder; channel base $d = 32$, latent dimension $k = 256$. Training follows (Fumero et al., 2026): Adam, learning rate 5×10^{-4} with a linear-decay schedule, weight decay 10^{-4}, MSE reconstruction, 500 epochs (we add early stopping on a 10% validation split).

Anchors. $N = k = 256$ images sampled class-balanced from the training split, shared across all seeds.

Latent dimension. The bottleneck $k = 256$ brackets two regimes. It sits well above the low-k band ($k \lesssim 16$) where (Fumero et al., 2026) report *memorization*, so the dynamics generalize and the attractors are non-trivial. It also sets the relative-representation dimension equal to the latent dimension, $N = k = 256$, following (Moschella et al., 2023)'s practice of matching the anchor count to the embedding dimensionality. On MNIST this suffices to reproduce both the $t = 0$ reproducibility and the breakdown.

Depth grid. $\mathcal{T} = \{0, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, \infty\}$.

Attractors and convergence. Following (Fumero et al., 2026), attractors are obtained by iterating the latent self-map $z_{t+1} = f_s(z_t) = E_s\big(D_s(z_t)\big)$ from $z_0 = E_s(x)$ (E_s, D_s frozen), using their stopping criterion $\|z_{t+1} - z_t\|_2^2 < 10^{-6}$ or a 3000-iteration cap. Two attractors are considered equal when their cosine similarity exceeds 0.99, as in (Fumero et al., 2026). This directional criterion is also the equivalence our relative representations induce: they read attractors only through cosine, so two on the same direction give identical codes (Appendix A).

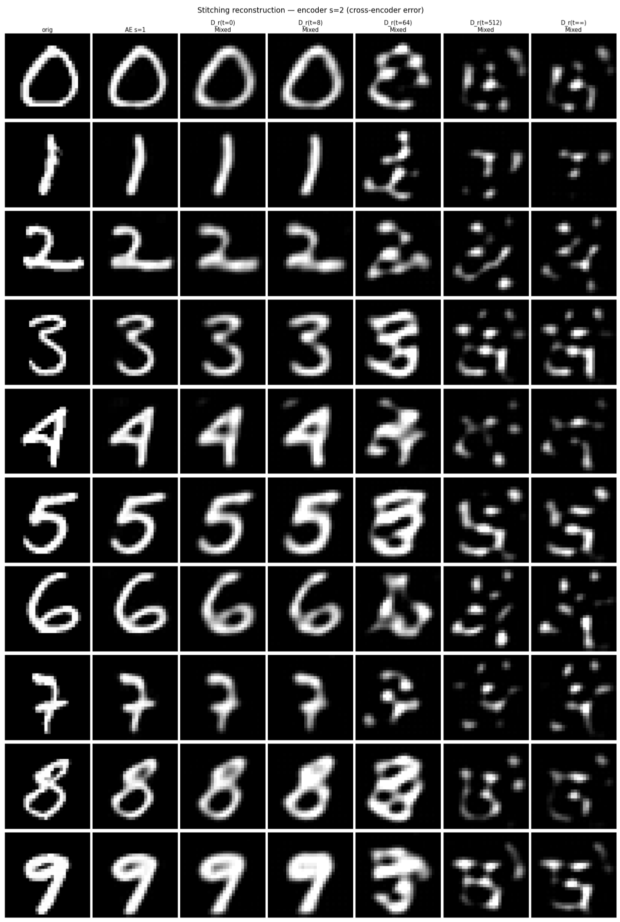
Stitching decoders. We use the zero-shot stitching of (Moschella et al., 2023): for each depth $t \in \{0, 8, 64, 512, \infty\}$ we train a *relative decoder* D_r^t on the reference seed's Mixed representation $\mathcal{M}^t$, with the reference encoder frozen. It reuses the autoencoder decoder above ($N = k$, so its linear input layer is unchanged), trained with the same optimizer and schedule.

Appendix D: Limitations and future work

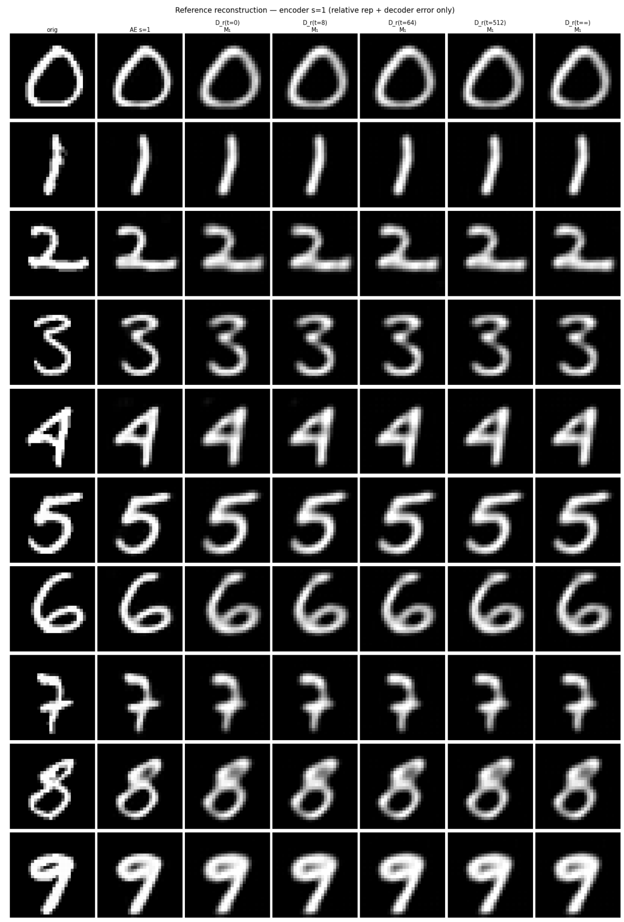
Limitations. Our claim joins a logical result with an empirical measurement. The result (Appendix A) holds for any models: shared attractor basins are *necessary* for Full alignment—the most permissive probe—so their failure is *sufficient* for non-alignment under any angle-preserving map. Specific to our setting are all our empirical findings, measured on one dataset (MNIST) and one architecture family (convolutional AEs); we cannot claim the breakdown beyond this regime. The per-anchor ordering is reproducible (split-half $r = 0.80$), hence real, but second-order beside the universal collapse, and its cause remains open.

Future work. (i) Test the breakdown on other datasets and architectures; all our experiments run once a latent vector field $f_s = E_s \circ D_s$ is defined. (ii) Explain the per-anchor ordering—what encoder-invariant property, beyond basin membership, drives it. (iii) Seek corrections beyond angle-preserving maps that recover the comparability of iterated representations.

Other relevant figures



(a) Stitched: $s = 2$ code through the reference decoder (cross-encoder).



(b) Reference control: $s = 1$ through its own decoder.

Figure 10. Stitching stops reconstructing the input. Decoder outputs at increasing depth t: cross-encoder stitching (a) no longer reconstructs the input by $t = \infty$, while the same-encoder reference control (b) still does at every depth—so the failure is the cross-model drift, not the relative code or the decoder.